



# Armada Kalamunda Group

## Head Injury Guideline

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### 1. Purpose

To guide staff with clinical decision-making in assessment of head injuries in Adult and Paediatric patients in the Emergency Department.

### 2. Applicability

This guideline has been developed to guide clinical decision-making for the medical, nursing and allied health staff within the Armada Kalamunda Group (AKG) Emergency Department.

### 3. Head Injury Guidelines

In **all** head injured patients consider the possibility of **cervical spine injury** ([link to guideline](#))

#### 3.1 Adults

- Refer to [RPH Head Injuries Clinical Guideline \(ED\)](#)
- [Algorithm: Initial Management of Adult Closed Head Injury](#)
- [Algorithm: Initial Management of Adult Mild Closed Head Injury](#)

#### 3.2 Paediatrics (<16 years old)

- Refer to [PCH Emergency Department Guidelines - Head Injury](#)
- [PREDICT Guideline for Children with Mild to Moderate Head Injuries](#)

If child < 2 years old, mandatory to complete Paediatric Injury Proforma

### 4. Resources

#### 4.1 Discharge Advice Sheets

All patients (or responsible adult/guardian) discharged from ED with a head injury should be given clear verbal and written head injury discharge advice and safety netting

- [WA Health Emergency Department Adult Discharge Advice Sheet - Head Injury](#)
- [PCH Health Facts - Head Injury](#)
- [PCH Keeping Our Mob Healthy - Head Injury](#)

#### 4.2 Concussion Resources

- [North Metropolitan Health Service - Concussion](#)
- [Concussion in Australian Sport \(concussioninsport.gov.au\)](#)
- [Sports Concussion Assessment Tool \(SCAT6\)](#)

### 5. Legislative context

The following documents are required to give effect to this policy:

- HDWA [Recognising and Responding to Acute Deterioration Policy MP 0171/22](#)
- HDWA [Consent to Treatment Policy MP 0175/22](#)

### 6. Supporting information

The following documents support the implementation of this policy:

- [Recognition and Response to Acute Deterioration Policy AKG-POL-0388-22](#)
- [Management of Patients with a Spinal Injury Procedure AKG-PRO-0627-24](#)

### 7. Compliance, monitoring and evaluation

Compliance with this policy will be monitored via:

- Monitoring of reported clinical incident data via Datix CIMS.

Department specific findings are to be reported at the departmental meeting for review and identification of any opportunities for improvement.

## 8. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 8 – Action 8.04 Recognising acute deterioration
- Standard 8 – Action 8.04 Escalating care

## 9. References

1. RPH Head Injury Clinical Guideline (ED):  
[https://healthpoint.hdwa.health.wa.gov.au/policies/Policies/EMHS/RPH/Head%20Injuries%20Clinical%20Guideline%20\(ED\).pdf](https://healthpoint.hdwa.health.wa.gov.au/policies/Policies/EMHS/RPH/Head%20Injuries%20Clinical%20Guideline%20(ED).pdf)
2. PCH ED Head Injury Guideline: <https://pch.health.wa.gov.au/For-health-professionals/Emergency-Department-Guidelines/Head-injury>
3. Initial management of closed head injury in adults: Summary document (nsw.gov.au):  
[https://aci.health.nsw.gov.au/\\_data/assets/pdf\\_file/0004/195151/NSW-Health-Initial-management-closed-head-injuries-summary.pdf](https://aci.health.nsw.gov.au/_data/assets/pdf_file/0004/195151/NSW-Health-Initial-management-closed-head-injuries-summary.pdf)

## 10. Document owner

Enquiries relating to this guideline may be directed to:

Title: Dr Emma Trace, Consultant Emergency Physician  
 Department: Emergency Department  
 Email: emma.trace@health.wa.gov.au

## 11. Review

This guideline will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

| Version | Effective from | Effective to | Amendment(s) |
|---------|----------------|--------------|--------------|
| V1.0    | 09/12/2024     | 09/12/2027   |              |

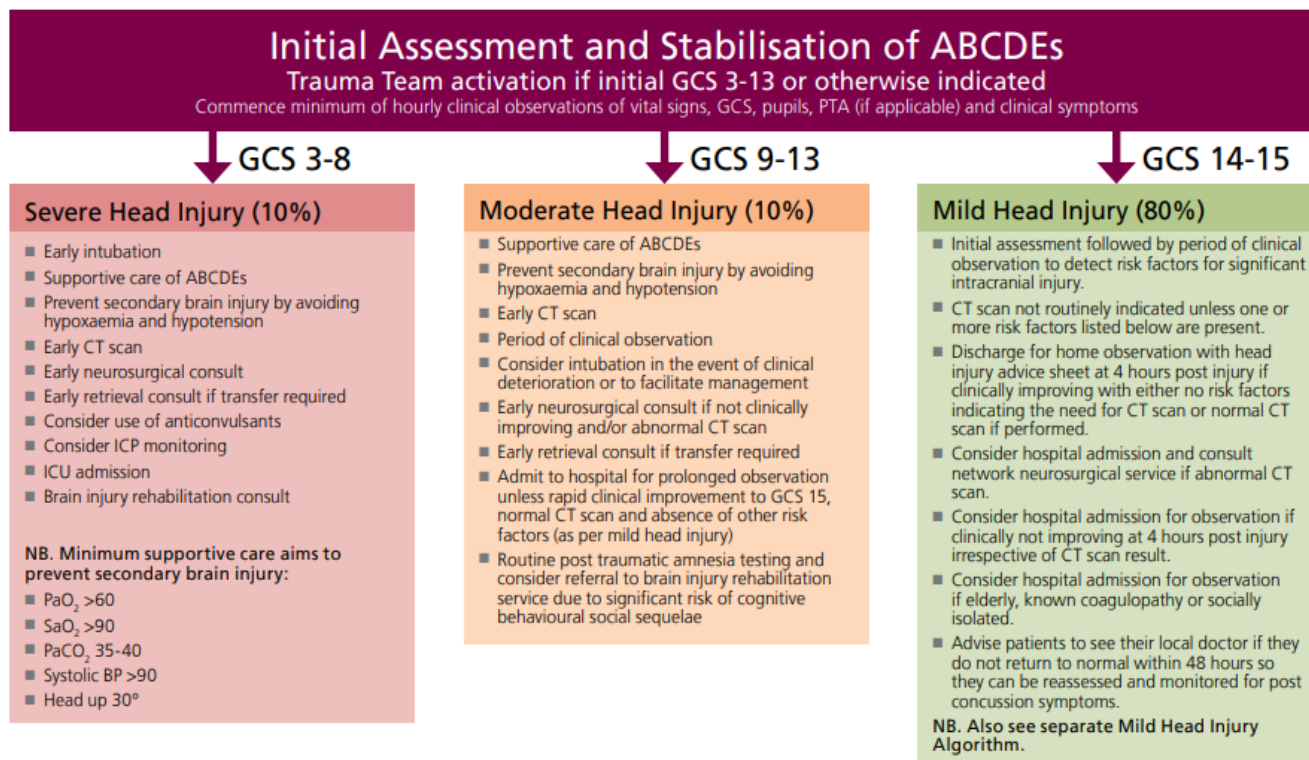
## 12. Authorisation

This guideline has been authorised and issued by the Director Clinical Services.

|                    |                                                    |
|--------------------|----------------------------------------------------|
| Authorised by      | Dr Tania Strickland – A/Director Clinical Services |
| Authorisation date | 02/12/2024                                         |
| Published date     | 09/12/2024                                         |

Appendix 1: Algorithm: Initial Management of Adult Closed Head Injury<sup>3</sup>

## Initial Management of Adult Closed Head Injury



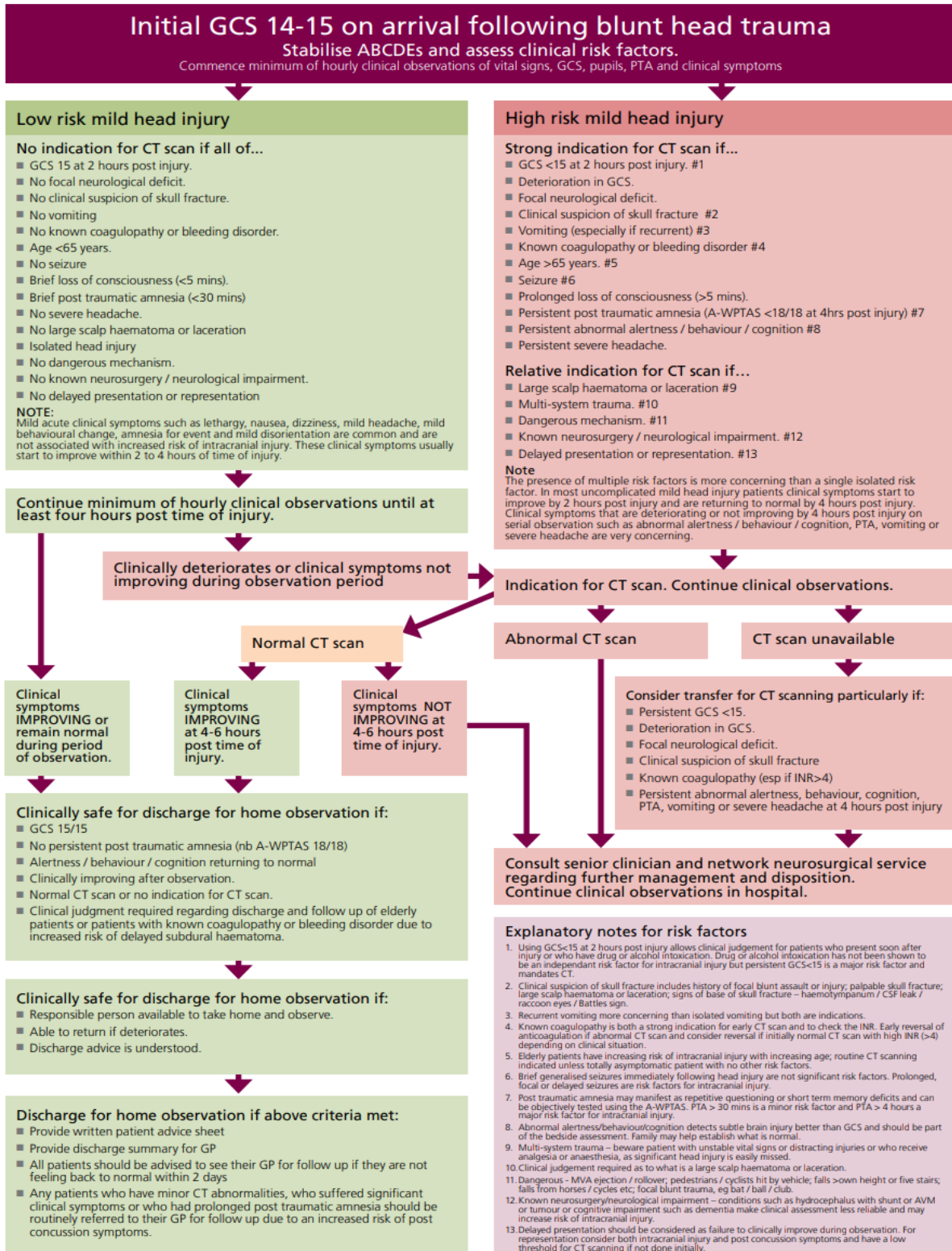
| Risk factors indicating potentially significant mild head injury                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>GCS &lt;15 at 2 hours post injury</li> <li>Deterioration in GCS</li> <li>Focal neurological deficit</li> <li>Clinical suspicion of skull fracture</li> <li>Vomiting (especially if recurrent)</li> <li>Known coagulopathy / bleeding disorder</li> </ul> | <ul style="list-style-type: none"> <li>Age &gt;65 years</li> <li>Post traumatic seizure</li> <li>Prolonged loss of consciousness (&gt;5 min).</li> <li>Persistent post traumatic amnesia (AWPTAS &lt;18/18)*</li> <li>Persistent abnormal alertness / behaviour / cognition*</li> <li>Persistent severe headache*</li> </ul> | <ul style="list-style-type: none"> <li>Large scalp haematoma or laceration.**</li> <li>Multi-system trauma**</li> <li>Dangerous mechanism**</li> <li>Known neurosurgery / neurological deficit.**</li> <li>Delayed presentation or representation**</li> </ul> |
| <p>* particularly if persists at 4 hours post time of injury<br/>           **clinical judgement required</p>                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                |

| What should be done when patients with closed head injury acutely deteriorate?                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Early signs of deterioration</b> <ul style="list-style-type: none"> <li>Confusion</li> <li>Agitation</li> <li>Drowsiness</li> <li>Vomiting</li> <li>Severe headache</li> </ul>                                                                                    | <b>Clinical approach</b> <ul style="list-style-type: none"> <li>Resuscitate ABCDEs and exclude non head injury cause</li> <li>Supportive care of ABCDEs</li> <li>Early intubation if indicated</li> <li>Immediate CT scan</li> <li>If clinical or CT evidence of raised ICP/mass effect consult with network neurosurgical and retrieval services re;               <ul style="list-style-type: none"> <li>- short term hyperventilation to PaCO<sub>2</sub> 30-35</li> <li>- bolus of mannitol (1g/kg)</li> <li>- local burr holes/craniectomy when more than 2 hours from neurosurgical care</li> <li>- prophylactic anti-convulsants</li> </ul> </li> </ul> |
| <b>Late signs of deterioration</b> <ul style="list-style-type: none"> <li>Decrease in GCS by two or more points</li> <li>Dilated pupil(s)</li> <li>Focal neurological deficit</li> <li>Seizure</li> <li>Cushing's response – bradycardia and hypertension</li> </ul> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

PTA = Post Traumatic Amnesia

Appendix 2: Algorithm: Initial Management of Adult Mild Closed Head Injury<sup>3</sup>

## Initial Management of Adult Mild Closed Head Injury



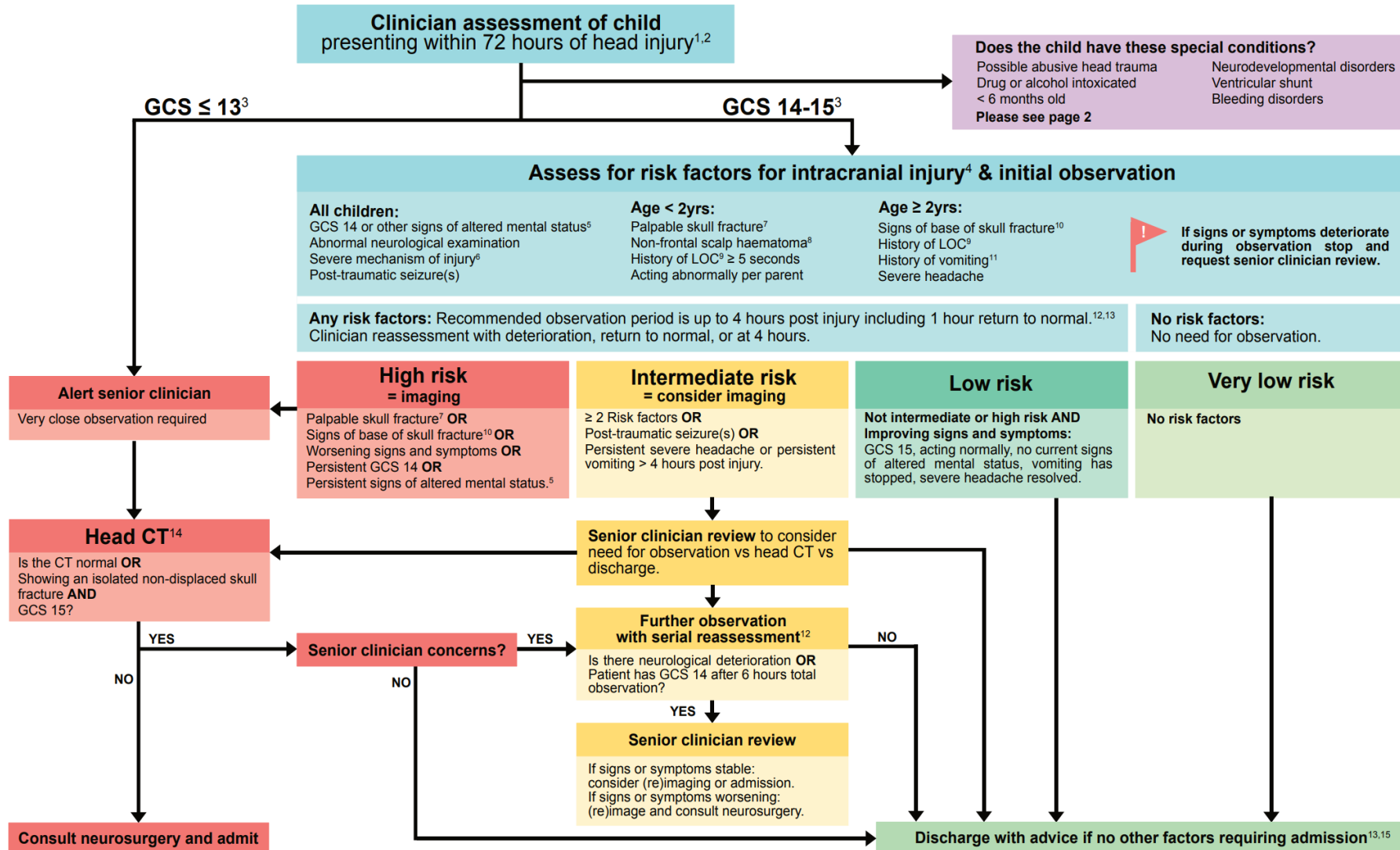
# Appendix 3: PREDICT Guideline



From: PREDICT GUIDELINE FOR CHILDREN WITH MILD TO MODERATE HEAD INJURIES (see [www.predict.org.au](http://www.predict.org.au)) Version 1.0 [150121]

## Algorithm: Imaging & Observation Decision-Making for Children with Head Injuries

Further details and footnotes are important to interpretation of the algorithm. Please see page 2.





## Further details to aid algorithm interpretation

- <sup>1</sup> Always consider possible cervical spine injuries and abusive head trauma in children presenting with head injuries.
- <sup>2</sup> Children with delayed initial presentation (24-72 hrs post head injury) and GCS 15 should be risk stratified the same way as children presenting within 24 hours. They do not need to be assessed with a further 4 hrs of observation.
- <sup>3</sup> Remember to use an age-appropriate Glasgow Coma Scale (GCS).
- <sup>4</sup> Risk factors adapted from Kuppermann N et al. *Lancet* 2009;374(9696):1160-70.
- <sup>5</sup> Other signs of altered mental status: agitation, drowsiness, repetitive questioning, slow response to verbal communication.
- <sup>6</sup> Severe mechanism of injury: motor vehicle accident with patient ejection or rollover, death of another passenger, pedestrian or cyclist without helmet struck by motor vehicle, falls of  $\geq 1\text{m}$  ( $< 2\text{ yrs}$ ), fall  $> 1.5\text{m}$  ( $\geq 2\text{ yrs}$ ), head struck by high impact object.
- <sup>7</sup> Palpable skull fracture: on palpation or possible on the basis of swelling or distortion of the scalp.
- <sup>8</sup> Non-frontal scalp haematoma: occipital, parietal, or temporal.
- <sup>9</sup> Loss of consciousness.
- <sup>10</sup> Signs of base of skull fracture: haemotympanum, 'raccoon' eyes, cerebrospinal fluid (CSF) otorrhoea or CSF rhinorrhoea, Battle's signs.
- <sup>11</sup> Isolated vomiting, without any other risk factors, is an uncommon presentation of clinically important traumatic brain injury (cTBI). Vomiting, regardless of the number or persistence of vomiting, in association with other risk factors, increases concern for cTBI.
- <sup>12</sup> Observation to occur in an optimal environment based on local resources. Frequency of observation to be  $\frac{1}{2}$  hourly for the first 2 hours, then 1-hourly until 4 hours post injury. After 4 hours, continue 2-hourly as long as the patient is in hospital. Observation duration may be modified based on patient and family variables. These include time elapsed since injury/symptoms and ability of child/parent to follow advice on when to return to hospital.
- <sup>13</sup> Shared decision-making between families and clinicians should be considered.
- <sup>14</sup> Do not use plain X-rays, or ultrasound of the skull, prior to or in lieu of CT scan, to diagnose or risk stratify a head injury for possible intracranial injuries.
- <sup>15</sup> Other factors warranting hospital admission may include other injuries or clinician concerns e.g. persistent vomiting, drug or alcohol intoxication, social factors, underlying medical conditions, possible abusive head trauma.



## Special Conditions

### Possible abusive head trauma



Follow local screening tools for abusive head trauma (AHT). CT should be used as initial diagnostic tool to evaluate possible intracranial injury and other injuries relevant for the evaluation of AHT e.g. skull fractures. The extent of the assessment of a child with possible AHT should be co-ordinated with the involvement of an expert in the evaluation of non-accidental injury.

### Drug or alcohol intoxicated



Treat as if the neurological findings are due to the head injury. Decision to CT scan or observe should be informed by risk factors for intracranial injury rather than the child being intoxicated.

### < 6 months of age



Consider at higher risk of intracranial injury with a lower threshold for observation or imaging. Discuss with a senior clinician.

### Neurodevelopmental disorders



It is unclear whether these children have a different background risk for intracranial injury. As these children may be difficult to assess, consider structured observation or head CT scan and include the paediatric team that knows the child (parents, caregivers, and clinicians) in shared decision-making.

### Ventricular shunt (e.g. ventriculo-peritoneal shunt)



Consider structured observation over immediate CT scan if there are no risk factors of intracranial injury. If there are local signs of shunt disconnection/shunt fracture (such as palpable disruption or swelling) or signs of shunt malfunction, consider obtaining a shunt series based on consultation with a neurosurgical service.

### Bleeding disorders or anti-coagulant or anti-platelet therapy



Urgently seek advice from the treating haematology team around risk of bleeding and management of coagulopathy. Consider structured observation over immediate CT scan if there are no risk factors for intracranial injury. If there is a risk factor for intracranial injury a head CT should be performed. If there is a deterioration in neurological status, perform urgent head CT scan.

### Coagulation factor deficiency

CT scan or decision to observe must not delay the urgent administration of replacement factor.

### Immune thrombocytopenias (ITP)

Check a platelet count in all patients and blood group in all symptomatic patients if not already available. For ITP with platelet counts  $< 20 \times 10^9/\text{L}$ , consider empirical treatment after discussion with the treating haematology team.

### On warfarin therapy or other newer anticoagulants (e.g. direct oral-anticoagulant) or anti-platelet therapy

Consider CT regardless of the presence or absence of risk factors for intracranial injury. Seek senior clinician review to inform timing of the CT and discuss the patient with the team managing the anticoagulation regarding early consideration of reversal agents. For children on anticoagulation therapy, if available, check the appropriate anticoagulant measure (e.g. International normalised ratio).

Citation: Babl FE, Tavender E, Dalziel S. On behalf of the Guideline Working Group for the Paediatric Research in Emergency Departments International Collaborative (PREDICT). Australian and New Zealand Guideline for Mild to Moderate Head injuries in Children – Algorithm (2021). PREDICT, Melbourne, Australia.